The mathematics and the debt

Companion to strips 01–08 · supplied structure, derived consequences, explicit limits

What these pictures can establish

This is a supplied classical model with one degree of freedom. It makes a local Hamiltonian construction and one Noether example visible. It does not derive mechanics from marks alone, nor yet explain quantum mechanics.

01–04 · Comparing, composing, refining, returning

We supply an affine plane, comparison by translation, and reversible rotations. Tip-to-tail composition gives vector addition. Strip 3 samples a prescribed differentiable path; drawings alone do not prove a smooth limit. Strip 4 introduces cyclic position (angular phase), not the specialized geometric phase of transport/holonomy, and not a quantum amplitude. It shows three intermediate directions and a return.

05 · The extra structure that makes Hamilton possible

We now supply a STATE plane with dimensionless conjugate coordinates q,p and oriented area $\omega = dq \wedge dp$. In strips 6–8 black measures q (right positive), green measures p (up positive); left/down are negative. The green tip locates the state relative to the black tail. These are state components, not spatial directions. Conjugacy is assumed, not established by a right angle. In physical units p dq has units of action. Momentum need not equal velocity.

05 → 06 · The local derivation

Let u,v be the two rates of change: a dot means change per unit time. Assume an autonomous C^1 rule and preservation of oriented area for EVERY small cell. Local expansion in one component must cancel that in the other: $\partial u/\partial q + \partial v/\partial p = 0$. A partial derivative is a slope while holding the other coordinate fixed. This cancellation lets one function H generate both rates by its two slopes. Formally, $\alpha = u\,dp - v\,dq$ is closed; on a simply connected local patch it equals dH. Thus

$$\dot{q} = u = \partial H/\partial p, \qquad \dot{p} = v = -\partial H/\partial q.$$

These are Hamilton’s equations. H is fixed up to an additive constant. A single patch in strip 5 illustrates the hypothesis, not its general proof. Topology can obstruct a global H; identifying H with physical energy needs a physical model.

06 · An exact turning rule

Choose $H = (q^2+p^2)/2$. The two rate panels use the SAME state (2.1,1.2), at the SAME instant. Orange lengths are 1.2 and 2.1 on a common rate scale. They are not successive kicks. The last panel joins each pair as in strip 2: three exact states at times 0, 0.5, 1 share one origin and scale; pale means earlier. The exact evolution is

$$q' = q\cos h + p\sin h, \qquad p' = p\cos h - q\sin h.$$

This preserves ω and H. The Euler assignment $(q + h\,p, p - h\,q)$ multiplies area and q^2+p^2 by $1+h^2$ instead. Circles belong to this chosen example and scaling, not every Hamiltonian system.

07 · A different rule and its continuous symmetry

Switch to $H = p^2/2$: $\Phi_t(q,p) = (q+t\,p, p)$. A configuration shift $T_a(q,p) = (q+a,p)$ is canonical and leaves H unchanged, so $\Phi_t T_a = T_a \Phi_t$. The two order panels both start at (2,0.65), use shift a=1.5 and time t=1, and finish at (4.15,0.65). Their pale intermediate q values are 3.5 and 2.65. This is a different rule: the oscillator does NOT have this translation symmetry.

08 · The Noether charge and why it is conserved

Translation is generated by $Y = \partial/\partial q$; its pairing with ω gives dp, so the charge is $J = p$. Since H has no q dependence, $dJ/dt = \dot{p} = -\partial H/\partial q = 0$. Equivalently, use $S = \int (p\,\dot{q} - H)\,dt$. Independent interior variations of q,p, with $\delta q = 0$ at time endpoints, give Hamilton’s equations. Constant q translations preserve the action and map endpoint data to translated data; they are not fixed-endpoint variations. Noether conservation holds along stationary trajectories. Equal drawn marks alone do not prove this theorem.

Limits and the remaining debt

Supplied: geometry, oriented pairing, regularity, the two model rules and variational hypotheses. Derived conditionally: local Hamiltonian form, exact example flows and conserved translation charge. Not derived: the earlier Genesis/L1 result, an identity of holonomy with Noether, quantization, Born weights, Everett branching or Bell correlations. Novice comprehension remains to be tested.